Telecom Customer Churn Prediction Using Machine Learning

Abstract

This project develops a machine learning model to predict whether a telecom customer is likely to discontinue a service. Customer demographics, billing details, contract information, and service usage are analyzed using Python and Scikit-learn. The workflow includes data cleaning, exploratory data analysis, model training, evaluation, and customer risk segmentation to help businesses reduce churn.

Introduction

Customer churn directly affects revenue and customer acquisition costs. Predicting churn early allows telecom companies to take preventive actions such as targeted offers, improved customer support, and loyalty programs.

Tools Used

Google Colab, Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and SHAP (optional).

Steps Involved in Building the Project

1. Collected the IBM Telco Customer Churn dataset.
2. Cleaned missing values, duplicates, and incorrect data types.
3. Performed exploratory data analysis using charts and summaries.
4. Selected features and converted categorical variables for machine learning.
5. Split the dataset into training and testing sets (80:20).
6. Trained a Logistic Regression model to predict churn.
7. Evaluated the model using Accuracy, Precision, Recall, F1-score, ROC-AUC, and a Confusion Matrix.
8. Generated customer risk scores and feature importance insights.
9. Suggested business strategies to reduce customer churn.

Expected Deliverables

A Python notebook, a trained churn prediction model, a CSV containing customer risk predictions, and supporting visualizations.

Conclusion

This project demonstrates how machine learning can support customer retention by identifying high-risk customers before they leave. The workflow combines data preprocessing, predictive modeling, evaluation, and explainable insights, making it suitable as a resume-worthy beginner-to-intermediate AI/ML project.